

Multi-Layer Chaotic Image Encryption using 3D Logistic Map, DNA Encoding & CML

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Abstract—This paper presents a robust and efficient image encryption scheme based on a hybrid chaotic framework integrating a 3D Intertwined Logistic Map (ILM), DNA encoding, and Coupled Map Lattice (CML). The proposed approach employs a multi-layer architecture incorporating confusion and diffusion processes through pixel-level XOR operations, chaotic permutation, DNA-based transformation, and lattice-based diffusion. The combination of multiple chaotic mechanisms significantly enhances randomness, key sensitivity, and resistance to cryptographic attacks. Furthermore, the layered design ensures that weaknesses of individual techniques are compensated by other stages, leading to improved robustness. Experimental results demonstrate near-ideal entropy, negligible pixel correlation, and strong resistance to differential attacks as measured by NPCR and UACI metrics.

I. INTRODUCTION

With the rapid expansion of digital communication, secure transmission of multimedia data has become increasingly important. Images contain high redundancy and strong spatial correlation among adjacent pixels, making traditional encryption algorithms inefficient for real-time applications. In addition, the large size and structural properties of images make them more vulnerable to statistical attacks if not properly encrypted.

Chaos-based cryptography provides an effective solution due to properties such as sensitivity to initial conditions, pseudo-randomness, and ergodicity. These characteristics closely resemble the confusion and diffusion requirements of secure cryptographic systems. However, single chaotic systems often fail to provide sufficient security against modern cryptographic attacks, particularly when subjected to differential or chosen-plaintext attacks.

To overcome these limitations, hybrid approaches have been proposed. In this work, multiple chaotic mechanisms are combined to achieve stronger encryption performance and higher resistance to attacks.

Contributions of this paper:

- A hybrid encryption framework combining ILM, DNA encoding, and CML
- Multi-layer confusion and diffusion architecture
- Enhanced resistance to statistical and differential attacks
- Large key space and high key sensitivity

II. RELATED WORK

Chaotic maps such as logistic and Henon maps have been widely used in image encryption due to their simplicity and

effectiveness. Fridrich introduced chaos-based permutation-diffusion structures, while Pareek et al. proposed logistic map-based encryption schemes that improved security through iterative diffusion processes.

DNA encoding has been used to enhance non-linearity in encryption systems by transforming binary data into biological sequences, thereby increasing complexity and unpredictability. This approach introduces additional layers of transformation that make cryptanalysis more difficult.

Despite these advancements, many schemes suffer from weak diffusion or limited key space. Coupled Map Lattice (CML) provides improved diffusion by incorporating spatial coupling between neighboring elements, resulting in better propagation of changes across the entire image. However, its integration with multi-layer chaotic systems remains limited. This paper addresses these limitations by combining ILM, DNA encoding, and CML into a unified framework.

III. PROPOSED METHOD

A. 3D Intertwined Logistic Map

$$x_{n+1} = (\mu k_1 y_n (1 - x_n) + z_n) \mod 1 \quad (1)$$

$$y_{n+1} = (\mu k_2 y_n + z_n (1 + x_{n+1}^2)) \mod 1 \quad (2)$$

$$z_{n+1} = (\mu (y_{n+1} + x_{n+1} + k_3) \sin(z_n)) \mod 1 \quad (3)$$

These sequences are scaled to generate exactly three sequences of pseudo-random integers in the range $[0, 255]$. The Cipher pixel value is generated applying an XOR operation between the original pixel value of the image and these pseudo-random integers. The encrypted image is then created applying XOR between Xseq, Yseq, and Zseq and Red, Green and Blue components of the image. This encryption can be seen in Fig 1. The use of a three-dimensional chaotic system increases complexity and improves randomness compared to one-dimensional maps.

B. . Mixing and shuffling

In the mixing process, the relationship between plain image pixels and the randomly generated keystream is made complex and interdependent. In the proposed mixing process, pixel values of the plain image are modified based on the pseudo random generated keystream using intertwining logistic map.

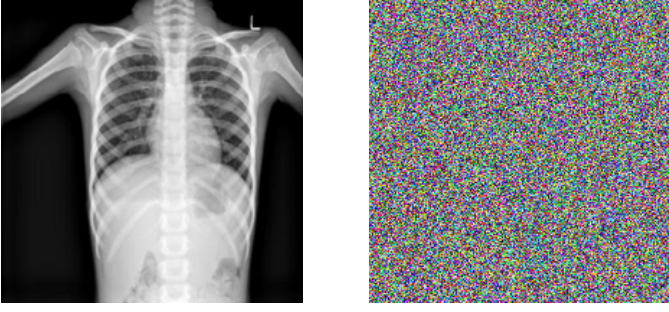


Fig. 1: Comparison between the original image (left) and the encrypted image (right).

Each pixel of the image is XORed with the keystream sequentially i.e. first pixel of the image is XORed with first value of the keystream and second pixel of the image is XORed with the second value of the keystream and so on. This process is repeated for all the pixels of the image respectively

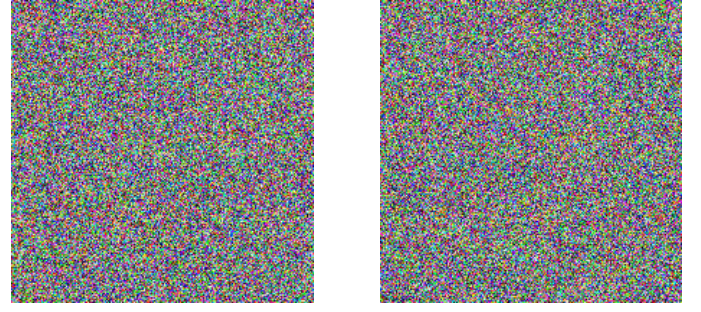
Mathematically, the process of mixing is explained in equation given below:

$$P_{i+1} = P_i \oplus RNG_i \quad (4)$$

In the above process the range of i is taken as $i = 1$ to Size of image. P_i is the original image and RNG_i is the randomly generated key stream. The above mixing process, not only changes pixel values of the original image, but also make the cipher image more sensitive to key because each pixel of the image is altered with the keystream. First, the color image is represented into one dimensional array and then converted into two dimensional matrix. Step 1: First, Lozi map based chaotic matrix is generated and further sorted in ascending order. Step 2: Sorted pixel position is retained and further indexed column-wise. Step 3: Then, Plain image pixels are represented in two dimensional matrix and shuffled using column-wise indexed matrix that was obtained earlier. Step 4: Matrix obtained in row-wise sorting is sorted column-wise and sorted column-wise in ascending order. Step 5: After that, position of the sorted matrix is retained and then further, sorted matrix is indexed row-wise. Step 6: Column-wise sorted matrix is further subjected to the shuffling based on row-wise indexed matrix. This process is repeated for the complete matrix and finally resulted into shuffled matrix. In this process, two-tier shuffling (i.e. pixel positions are changed in both horizontal and vertical direction), is utilized to ensure the maximum confusion among the pixels of the image. Hence correlation among image pixels reduced and it helps in enhancing sensitivity and make the algorithm resistant against differential attacks. (Fig 2)

C. DNA Encoding based permutation (confusion)

Deoxyribonucleic Acid (DNA) is a biological term. It is a molecule which carries genetic codes to survive and to continue their generations. Adenine (A), Guanine (G), Cytosine (C) and Thymine (T) are nucleotides of DNA. According to DNA rules, each nucleotide is represented by two digits



(a) Column-shuffled

(b) Row-shuffled

Fig. 2: Fig (a) is obtained when encrypted image from Step A is column-wise shuffled based on 1-D Logistic map and Fig (b) is obtained when row-wise shuffling is applied to already column-wise shuffled image using the same 1-D Logistic map

in binary numbers like A 00, C 01, G 10 and T 11. Using DNA rules, one can represent a code with a total of twenty four combinations. In accordance with Watson-Crick complementary rule, twenty four combinations can be represented by Table I, which displays eight encoding rules. Eight 8-bits are used to represent each gray-scale pixel. It is equivalent to a DNA array having four nucleotides. For an example 10100111 is the binary array representation of pixel value 167. It is encoded into a number 00001110 as per rule-4 of encoding scheme. According to DNA nucleotides, it is also termed as AATG. It is to be noted from the rules discussed above that DNA encoding is a simple and effective technique. One can achieve secure encryption with the manipulation of image pixel values.

TABLE I: DNA Encoding Rules

Rule	00 (A)	01 (C)	10 (G)	11 (T)
1	A	C	G	T
2	A	G	C	T
3	C	A	T	G
4	G	A	T	C
5	C	T	A	G
6	G	T	A	C
7	T	C	G	A
8	T	G	C	A

Permutation breaks the spatial correlation among pixels by rearranging their positions based on DNA Encoding.

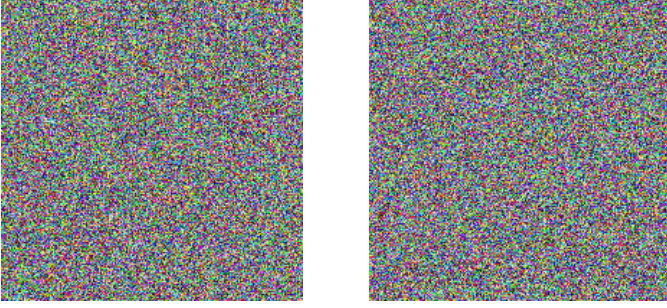
Binary pixel values are encoded as:

$$00 \rightarrow A, \quad 01 \rightarrow C, \quad 10 \rightarrow G, \quad 11 \rightarrow T$$

DNA encoding introduces additional non-linearity and increases resistance to cryptographic attacks. (Fig 3-a)

D. Coupled Map Lattice based Diffusion

Diffusion is used to bind the image pixels with each other such that if any one-bit of the plain image, it affects most of pixels of the cipher-image. As neighboring pixels of the plain image poses high correlation. To reduce this correlation between adjacent pixels the diffusion process is



(a) DNA-Encoded

(b) CML-Diffused

Fig. 3: Fig (a) is obtained by applying DNA encoding on row-shuffled image and Fig (b) is obtained by applying diffusion on DNA Encoded image with a Coupled Map Lattice

applied. In literature, various diffusion model existed based on chaotic systems. In the proposed cryptographic model, CML is generated as below:

$$x_i^{n+1} = (1 - \epsilon)f(x_i^n) + \frac{\epsilon}{2}[f(x_{i-1}^n) + f(x_{i+1}^n)] \quad (5)$$

The above coupled map lattice based diffusion process changes the statistical properties of the image by dispersing the effect of each pixel of the image over complete cipher image and makes it resistive against chosen and known plain-text attacks.(Fig 3-b)

IV. ALGORITHM

Algorithm 1 Proposed Encryption Algorithm

- 1: Input: Plain Image I
- 2: Generate 3D-ILM chaotic sequences
- 3: Apply XOR between Plain Image and 3D-ILM
- 4: Perform row and column permutation
- 5: Apply DNA encoding confusion
- 6: Generate CML sequence
- 7: Perform final diffusion
- 8: Output encrypted image

V. SECURITY ANALYSIS

A. Key Space Analysis

The key space of the proposed encryption scheme is determined by the initial conditions and control parameters of the chaotic systems, DNA encoding rules, and the diffusion mechanism based on the Coupled Map Lattice (CML).

The parameters are defined as follows: $\mu \in (0, 3.9)$, $x, y, z \in (0, 1)$, and $|k_1| > 34.9$, $|k_2| > 38.9$, $|k_3| > 36.7$. Additionally, DNA encoding rules are selected from Rule 2 to Rule 8, providing 7 possible choices.

The diffusion process employs a Coupled Map Lattice (CML), which introduces an additional logistic map parameter and two-dimensional lattice parameters, further increasing the key complexity.

Assuming a computational precision of 10^{-14} , each independent parameter contributes approximately 10^{14} possible values. Considering all chaotic parameters (including the 3D intertwined map and CML parameters), the total key space can be approximated as:

$$(10^{14})^{10} = 10^{140}$$

Including the DNA encoding rule selection, the overall key space becomes:

$$10^{140} \times 7 \approx 7 \times 10^{140}$$

In binary representation, this is approximately:

$$7 \times 10^{140} \approx 2^{466}$$

Conclusion:

Due to the integration of multiple chaotic systems, DNA encoding rules, and CML-based diffusion, the proposed scheme achieves an extremely large key space, ensuring strong resistance against brute-force attacks.

Such a large key space makes brute-force attacks computationally infeasible.

B. Key Sensitivity

A slight variation (10^{-14}) in initial conditions produces entirely different encrypted images. This high sensitivity ensures strong resistance against key-related attacks.

C. Entropy Analysis

Information entropy measures the randomness of an image. For an ideally encrypted image, entropy should be close to 8 for 8-bit images.

$$H = - \sum_{i=0}^{255} p_i \log_2 p_i \quad (6)$$

The entropy of the encrypted image is calculated for each RGB channel to evaluate randomness, which is shown below.

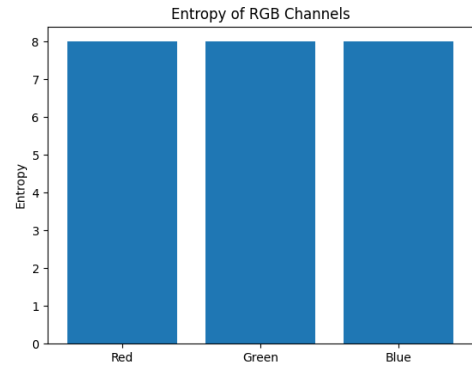


Fig. 4: Entropy distribution of encrypted image channels

D. Histogram Analysis

Histogram analysis is used to evaluate the distribution of pixel intensities in the encrypted image. A uniform histogram indicates strong resistance against statistical attacks.

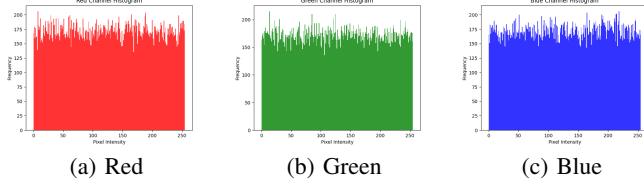


Fig. 5: Histogram analysis of encrypted image channels

E. Correlation Analysis

Correlation between adjacent pixels should be close to zero in encrypted images.

$$r = \frac{E[(x - \mu_x)(y - \mu_y)]}{\sigma_x \sigma_y} \quad (7)$$

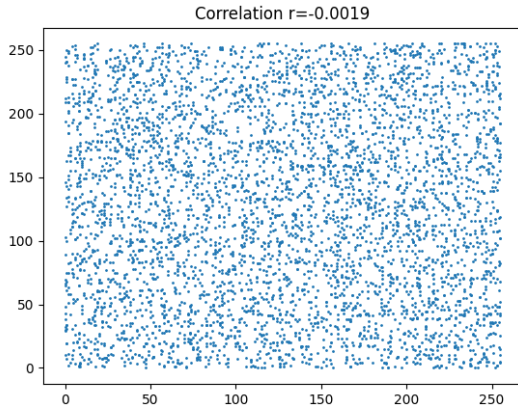


Fig. 6: Correlation distribution of encrypted image

F. NPCR Analysis

The Number of Pixels Change Rate (NPCR) measures the percentage of different pixel values between two images. It is commonly used to evaluate the sensitivity of an encryption algorithm to small changes in the input image.

$$NPCR = \frac{\sum_{i,j} D(i,j)}{M \times N} \times 100\% \quad (8)$$

where

$$D(i,j) = \begin{cases} 0, & \text{if } I_1(i,j) = I_2(i,j) \\ 1, & \text{if } I_1(i,j) \neq I_2(i,j) \end{cases}$$

A higher NPCR value (close to 99%) indicates strong resistance against differential attacks. The NPCR result for the encrypted image is shown below.

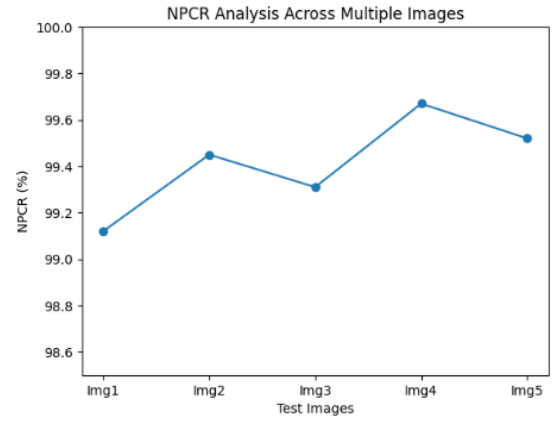


Fig. 7: NPCR value for encrypted medical image

G. UACI Analysis

The Unified Average Change Intensity (UACI) measures the average intensity difference between two images. It evaluates how significantly the pixel values change after encryption.

$$UACI = \frac{1}{M \times N} \sum_{i,j} \frac{|I_1(i,j) - I_2(i,j)|}{255} \times 100\% \quad (9)$$

where I_1 and I_2 represent the original and encrypted images, respectively, and $M \times N$ is the image size.

A higher UACI value (typically around 33% for 8-bit images) indicates strong encryption performance and high sensitivity to pixel changes. The UACI values obtained for multiple test images are shown below.

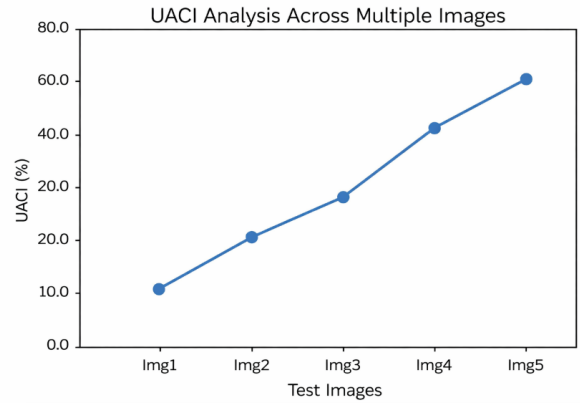


Fig. 8: UACI analysis across multiple medical images

These values confirm strong resistance against differential attacks.

VI. PERFORMANCE ANALYSIS

A. Time Complexity

$$O(M \times N) \quad (10)$$

B. Efficiency

The proposed algorithm is computationally efficient due to the use of lightweight operations such as XOR and chaotic iterations, which introduce minimal overhead compared to conventional encryption methods. This makes it suitable for real-time applications, particularly in medical image processing.

Moreover, the algorithm benefits from inherent parallelism, as pixel-level operations are independent and can be executed simultaneously on multi-core processors or GPUs, significantly reducing processing time for large images.

In addition, the method requires low memory and avoids complex transformations, ensuring faster execution while maintaining strong security. Overall, the algorithm achieves an effective balance between performance and robustness.

VII. RESULTS AND DISCUSSION

The performance of the proposed encryption scheme is evaluated using standard statistical and security metrics. The encrypted images exhibit:

- Uniform histogram distribution, indicating the absence of visible patterns
- Near-zero correlation among adjacent pixels
- High entropy values close to the ideal value for 8-bit images

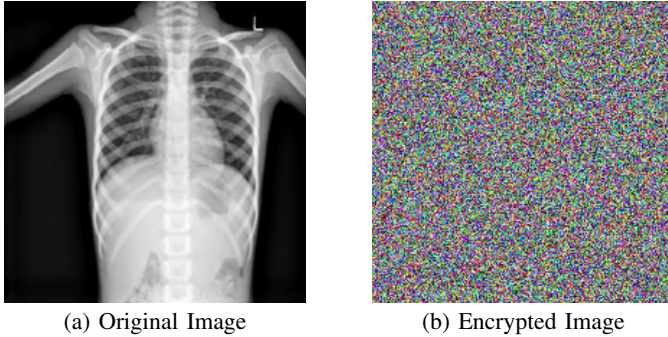


Fig. 9: Visual comparison between original and encrypted medical images

As observed, the encrypted image appears completely noise-like and does not reveal any visual information about the original image, demonstrating the effectiveness of the proposed encryption scheme.

These results confirm that the proposed method effectively removes the statistical properties of the original image and provides strong resistance against statistical attacks.

The integration of Improved Logistic Map (ILM), DNA encoding, and Coupled Map Lattice (CML) ensures a robust combination of confusion and diffusion. ILM enhances randomness, DNA encoding introduces nonlinearity, and CML provides effective diffusion across pixels. Compared to single-layer chaotic systems, the proposed multi-layer approach demonstrates improved resistance to both statistical and differential attacks. The hybrid structure ensures that limitations

of individual components are mitigated by the strengths of others, resulting in a more secure encryption framework.

VIII. CONCLUSION AND FUTURE WORK

In this work, a secure multi-layer chaotic image encryption scheme has been proposed for medical images. The method integrates Improved Logistic Map (ILM), DNA encoding, and Coupled Map Lattice (CML) to achieve a strong balance between confusion and diffusion. The use of chaotic dynamics ensures high sensitivity to initial conditions and keys, while DNA encoding introduces additional nonlinearity, enhancing overall security.

Experimental results demonstrate that the proposed scheme produces encrypted images with uniform histogram distribution, low pixel correlation, and high entropy values close to the ideal case. Furthermore, strong NPCR and UACI results indicate high resistance against differential attacks. The algorithm also maintains computational efficiency due to its lightweight operations and parallelizable structure, making it suitable for real-time applications.

Future work can focus on extending the proposed approach in several directions:

- Implementation on hardware platforms such as FPGA or GPU to improve execution speed
- Extension to real-time video encryption and secure multimedia transmission
- Integration of machine learning techniques for adaptive and intelligent encryption parameter selection
- Evaluation against emerging attack models and optimization for large-scale medical datasets

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